

Mohsen Shahverdi

Senior AI Engineer | Agent Systems & Infrastructure | Production LLM Platforms

Heidelberg, Germany | +49 176 71936945 | mohsen.shahverdi97@gmail.com | linkedin.com/in/mohsen-shahverdi/

/ SUMMARY

- Senior AI Engineer with **8+ years of experience** designing, deploying, and scaling production AI, backend, and full-stack systems for enterprise products.
- Specialize in **agent runtimes, AI compilers, multi-agent orchestration, LLM observability, context-window engineering, model routing, and production evaluation**.
- Translate ambiguous business requirements into reusable platform capabilities and deliver **integrated, testable enterprise applications** from specification through production.

/ PROFESSIONAL EXPERIENCE

Senior AI Engineer - Agentic AI Systems

Sep 2025 - Present

Lensing AI (formerly Everest Systems GmbH) - Heidelberg, Germany | AISpecify

- Core engineer on **AISpecify**, working within the Build/Compiler team to architect agent systems for a spec-driven platform that generates production-ready, ERP-scale enterprise applications.
- Architected and implemented the **AI compiler and code-generation pipeline**, coordinating specialized agents, tool execution, models, validators, repair stages, and MCP-based enterprise integrations.
- Improved the builder from a **five-page application with ~30% functional completion in one hour** to a **fully integrated 40-page enterprise application in ~25 minutes**, reducing generation cost by **~9x**; the architecture supports **200+ page** platforms.
- Designed and built an **AI observability layer** tracing every LLM call, agent execution, tool invocation, model decision, latency, failure, and orchestration step, making the pipeline explainable and debuggable in production.
- Designed and built a **graph-based orchestration architecture** that models dependencies among agents, tools, generation stages, and application artifacts, enabling dynamic task ordering, execution planning, and model/tool selection.
- Engineered **context-window management** with graph-driven context selection, targeted artifact and business-context injection, and conversation state to improve relevance for each LLM call.
- Built a flexible **plugin-based agent harness** and **agent-loop generation foundation**, allowing teams to compose reusable tools, validators, context providers, model strategies, and iterative agent workflows with minimal core-runtime changes.
- Designed and implemented **model-routing, parallel execution, evaluation, validation, consistency checking, and automated self-repair**; routing, parallelization, and context/prompt caching reduced platform processing costs by up to **60%**.

AI Software Engineer

Aug 2024 - Sep 2025 | 1 yr 2 mos

Hich Co - Birmingham, UK (Remote)

- Designed an event-driven recommendation platform using **Kafka, ClickHouse, collaborative filtering, content-based ranking, and clustering**, increasing user activity by **18%**, reducing post skipping by **33%**, and increasing average usage by **13%**.
- Developed a **semantic retrieval layer** over platform content to improve search, personalization, and LLM-assisted content generation workflows.
- Built fraud detection and user legitimacy scoring systems, improving content quality accuracy by **43%** and increasing monthly revenue by over **60%**.
- Fine-tuned **LLaVA-1.6 Vicuna-7B** with **LoRA** on **1.7M moderator-labeled samples** for automated content quality scoring, achieving **98.1% accuracy**.
- Developed OpenAI-powered content generation pipelines integrated with external news sources, increasing user interactions by **14%** and average session time by **8%**.
- Implemented Grafana monitoring for model quality, engagement metrics, system behavior, and business KPIs in production.

Machine Learning Engineer

Jul 2022 - Aug 2024 | 2 yrs 2 mos

Offerland - Vancouver, Canada (Remote)

- Worked directly with customers and farmland valuation specialists to understand existing evaluation workflows, domain expectations, and decision criteria before translating them into scalable ML products.
- Developed OfferFarm, a farmland price prediction platform using ExtraTrees, XGBoost, and Random Forest, achieving **94% accuracy** by combining regional, geographic, soil, and market-trend signals.
- Translated customer feedback into valuation improvements, including soil and location adjustments, kernel ridge regression, precise arable-acre computation through Google APIs, and better comparable-property selection.

OFFERLAND – CONTINUED

- Built concept-drift detection, automated retraining triggers, operational alerts, and Metabase dashboards for model performance, pricing trends, customer activity, and usage metrics.
- Reduced API latency by over **60%** through asynchronous requests and Redis caching.
- Engineered Offervista rental-trend pipelines covering **93%+** of Canadian postal codes.

Computer Vision Engineer / Technical Lead

Sep 2021 - Jul 2022 | 11 mos

Forsat Co - Tehran, Iran

- Led the computer vision workstream and worked directly with customers to understand operational constraints, install solutions in customer environments, and support production rollout on customer machines.
- Led data collection and augmentation, expanding a **300K-image** signature dataset into **2M** diverse samples.
- Developed a cheque signature detection pipeline using YOLOv5, PyTorch, and a custom autoencoder, achieving **96% mAP**.
- Implemented SigNet-based forgery detection with **92.7% accuracy** and deployed FastAPI inference services adapted to customer infrastructure.

Data Scientist

Jan 2021 - Dec 2021 | 1 yr

Bale Messenger - Tehran, Iran

- Launched a channel recommendation system, increasing channel joins by **41%** and average in-app time to **8 minutes**; fine-tuned ParsBERT for a banking QA chatbot, achieving **92% accuracy**.

Data Scientist

Mar 2020 - Jan 2021 | 11 mos

Yektanet - Tehran, Iran

- Implemented recommendation systems increasing conversion rate by **58%** and association-rule product bundling algorithms increasing CTR by **25%**.

Software Engineer

Mar 2018 - Dec 2019 | 1 yr 10 mos

ShaniKala - Tehran, Iran

- Developed backend features, customer-facing APIs, marketplace integrations, and recommendation features using ASP.NET, .NET Core, MSSQL, and NopCommerce, increasing similar-product CTR by **37%**.

/ SKILLS

Agentic AI & LLM Systems	Multi-Agent Orchestration, Agent Runtimes, AI Compiler Workflows, MCP, Tool Calling, LLM Observability, Context-Window Engineering, Dependency Graphs, Context Injection, Model Routing, Plugin Architectures, RAG, Evaluation, Guardrails
LLM Training & ML	Post-Training, Fine-Tuning, LoRA/PEFT, LLM Evaluation, PyTorch, Hugging Face, TensorFlow, scikit-learn, XGBoost, NLP, 3D Cooperative Perception, Computer Vision, Recommendation Systems, Time Series
Software Engineering	TypeScript, Node.js, Python, FastAPI, Flask, REST APIs, Microservices, Distributed Systems, System Design, Unit & E2E Testing, CI/CD, Git, GitLab
Data & Infrastructure	PostgreSQL, MySQL, SQL, ClickHouse, MongoDB, Redis, Kafka, PySpark, Docker, AWS, Grafana, Metabase, Tableau

/ EDUCATION

Master of Engineering - Artificial Intelligence Focus

Sep 2022 - Jun 2025

Politecnico di Milano - Milan, Italy | Telecommunication Engineering

Relevant coursework: Machine Learning, Deep Learning, Advanced Computer Vision, NLP, Advanced Signal Processing and Learning, Data Mining, Uncertainty in AI, Recommender Systems, Streaming Data Analysis.

Bachelor of Science - Computer Engineering

Sep 2015 - Feb 2020

Shahid Beheshti University - Tehran, Iran

Relevant coursework: AI, Computer Vision, Computational Intelligence, Data Mining, Data Structures, Algorithms, Signal Processing, Statistics, Calculus.